

Temporary Overvoltages: Part I

Leonardo Gabriel Granda Erazo
Universidad Nacional de Loja
Electrical Engineering
leonardo.granda@unl.edu.ec

Abstract—Overvoltages in electrical systems are very common events and occur at both high and low frequency; their classification depends on the underlying cause. This work analyzes temporary overvoltages, which are low-frequency and can be caused by load losses or ground faults. Those causes will be addressed in future work; here only the influence of distance and loads on the electrical system is analyzed, using a voltage profile. Matlab and Simulink were used for this purpose.

Index Terms—Temporary overvoltages, ground faults, load losses, voltage profile, Matlab.

I. INTRODUCTION

The Ecuadorian electrical system is organized into three main energy stages—generation, transmission, and distribution—which support one another to form an efficient and secure system. Good service exists when energy interruptions are minimal and customers receive the agreed voltage level. However, the longer a supply line, the higher the probability of a considerable voltage drop, which would affect service quality.

Although methods exist to keep the voltage drop in the most remote zone to a minimum, this work focuses on visualizing the voltage drop along a supply line within the EERSSA concession area. A solution to this problem will be modeled in later work.

II. PROBLEM STATEMENT

Using the EERSSA geographic information system, select a substation and a feeder to represent a distribution system designed to deliver energy to small towns and/or medium-sized industries through an overhead network.

The transformer rating is chosen according to the installed power of the feeder under study (a minimum of 10 MVA of installed load), taking the installed power to be 60% of the transformer capacity. The substation is assumed to be supplied from a 1 GVA generator with $X/R = 5$, representing the upstream electrical system.

At least four load subdivisions are placed along the feeder, representing the towns or industries supplied. If the feeder has a voltage regulator along the main branch, it must be simulated as an additional element to the four load subdivisions; for simplicity, it is represented through a capacitor bank accounting for 80% of the total reactive load downstream of its coupling point. If the geographic system provides the power data, the given value is used. The voltage profile is then shown at the different feeder nodes—including the transformer primary and secondary voltages—as a plot of distance versus RMS voltage in per unit (Fig. 1).

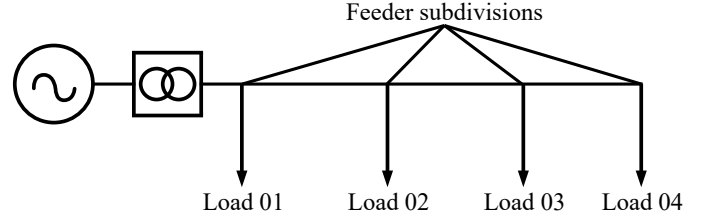


Fig. 1. System to be implemented in Simulink/Matlab.

III. STRUCTURE OF THE SYSTEM TO BE SIMULATED

After reviewing the EERSSA geographic information system, the system chosen was the Yanzatza substation, whose main feeder is the Paquisha feeder. Fig. 2 shows the overhead voltage section, used to estimate its length.

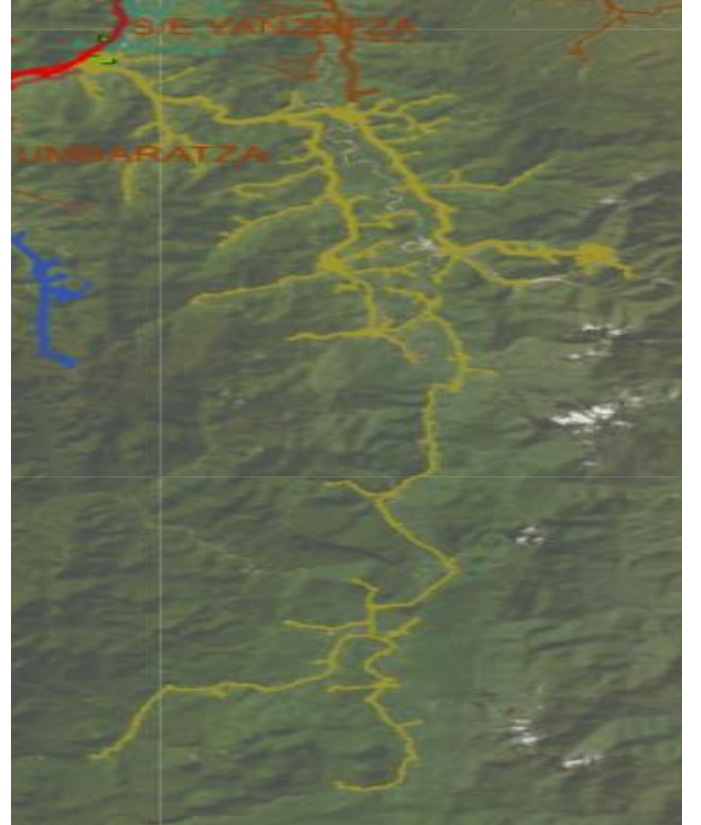


Fig. 2. Supply line of the Yanzatza substation.

Per the problem specifications, the substation is simulated

with a 1 GVA capacity; the installed power (the total load it supports) still had to be determined, which the site's tools allow to be approximated. Selecting the entire area covered by this supply line gives Fig. 3, from which the installed capacity of the substation is approximately 12.47 MVA. Since these are approximations, the next-higher round value was used in the simulation—a load power of 13 MVA.

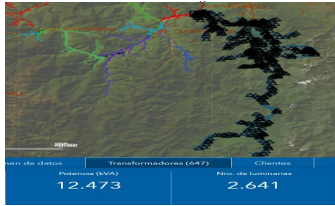


Fig. 3. Installed power in the selected area.

This section is three-phase, positive-sequence, at 22 kV, and belongs to the canton of Centinela del Cóndor in the province of Zamora Chinchipe. The line must therefore guarantee 22 kV throughout its coverage area, i.e., downstream of the main transformer the voltage level is 22 kV. From [1], [2], the sub-transmission lines and the Yanzatza substation are connected at 69 kV, so the main transformer converts the 69 kV coming from the transmission stage to 22 kV for distribution. With the overhead section and its power selected, the zones are then distributed.

A. Zone Breakdown

From Fig. 3, the line is divided into zones as shown in Fig. 4. Each zone and its specifications are detailed below.

Cross-referencing with Google Maps, zone 1 (bounded by the red curve in Fig. 4) mainly contains the town of Zumbi, with the rest being branches reaching down to Nangipe Bajo. Zone 2 (green curve) comprises Paquisha, El Dorado, La Teinza, La Wintza, Nuevo Quito, and Santa Rosa. Zone 3 covers the canton of Guayzimi, Congüime, Chinapintza, La Herradura, La Panguí, Alto Mayaycu, and Zurmi. Zone 4 begins at Pachikutza and runs through Las Orquídeas, Miazí, Shaime, Shamatak, Nuevo Paraíso, and Yawi. These data were obtained from observing the selected zones in Google Maps.

B. Zone Parameters

Following the same procedure used for the whole line, each zone was selected individually to establish an approximate load. The zone length was also measured, considering only a main branch and omitting the ramifications.

1) *Zone 1*: Figs. 5 and 6 show the power and length of this zone. It has a transformer power of 2437 kVA and 976 luminaires totaling 107145 W (about 107 kW), supplying an average of 1166 customers. Its main town is Zumbi, with branches down to Nangipe Bajo. For the simulation, a power of 2.5 MVA and a length of 14.2 km were adopted.

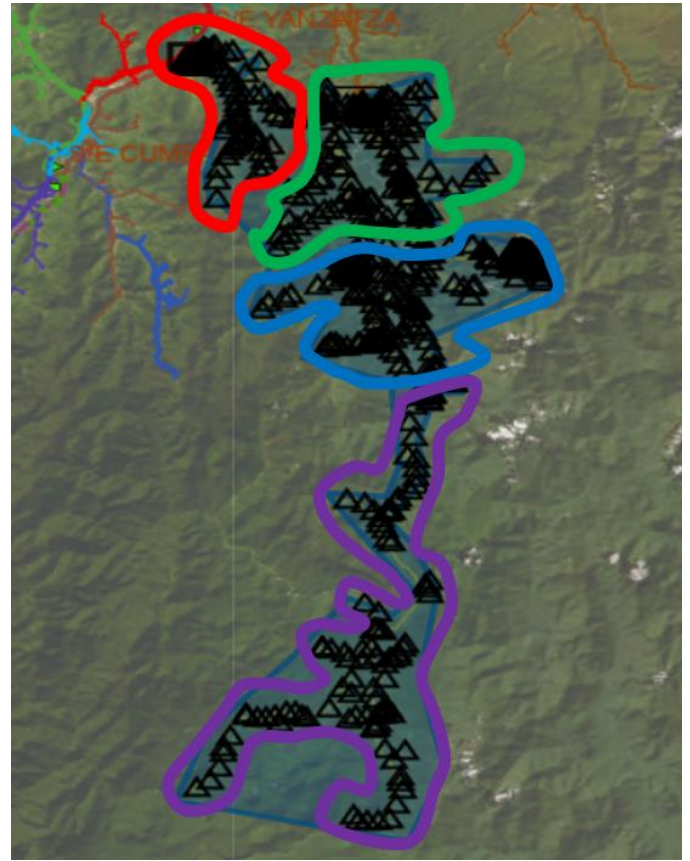


Fig. 4. Sectioning of the selected line.

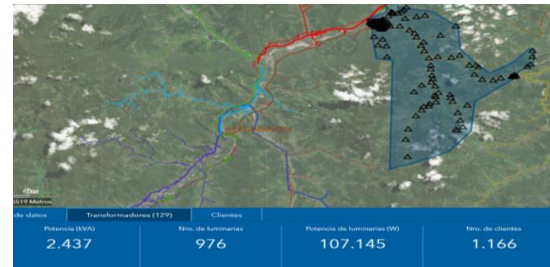


Fig. 5. Installed power in zone 1.

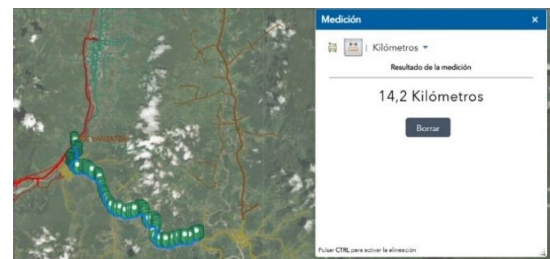


Fig. 6. Length of zone 1.

2) *Zone 2*: Figs. 7 and 8 show the data. This zone has an estimated transformer power of 2864 kVA and 729 luminaires totaling 89851 W (about 90 kW), supplying an average of 1223 customers, including Paquisha, El Dorado, Nuevo Quito, and Santa Rosa. A power of 3 MVA and a length of 10.2 km were adopted.

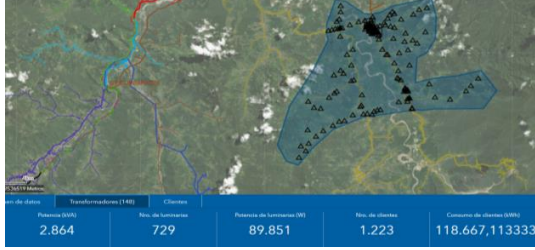


Fig. 7. Installed power in zone 2.

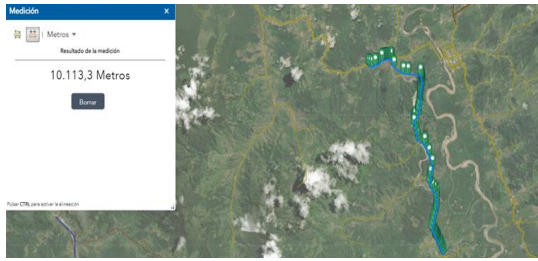


Fig. 8. Length of zone 2.

3) *Zone 3*: Figs. 9 and 10 show the data. Zone 3 has an estimated transformer power of 6071 kVA and 832 luminaires totaling 104890 W (about 105 kW), supplying an average of 1827 customers—the highest installed load of all zones. It feeds Guayzimi, Chinapintza, La Pangui, Alto Mayaycu, Zurmi, among others. Rounding up, a power of 6.1 MVA and a length of 20.7 km were adopted.

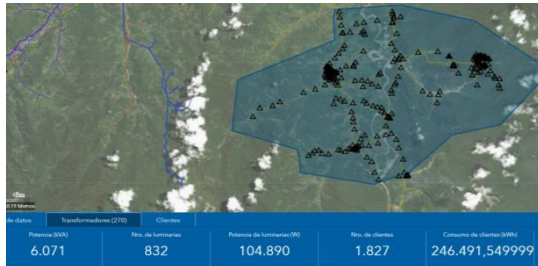


Fig. 9. Installed power in zone 3.

4) *Zone 4*: Figs. 11 and 12 show the data. Zone 4 has an average transformer power of 1329 kVA and 158 luminaires totaling 17440 W (17.4 kW), supplying an average of 597 customers—the lowest installed load but the longest distance. Its coverage begins at Pachikutza and runs through Las Orquídeas, Nuevo Paraíso, ending at Yawi. Rounding up, a power of 1.5 MVA and a length of 49.1 km were adopted.

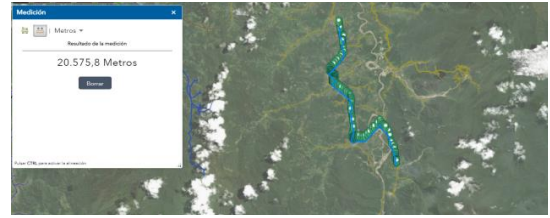


Fig. 10. Length of zone 3.

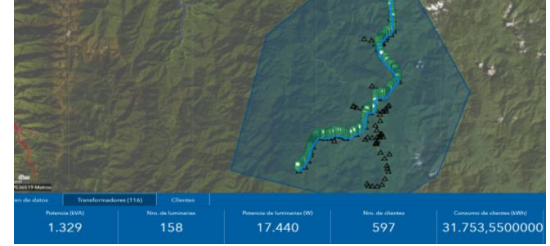


Fig. 11. Installed power in zone 4.

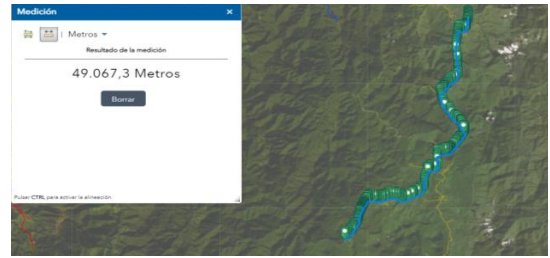


Fig. 12. Length of zone 4.

C. Simulation Characteristics

With the data collection complete, the values are summarized for the simulation.

1) *Transformer*: Table I shows the data of the main transformer used.

TABLE I
TRANSFORMER DATA.

Parameter	Value
Transformer power	21.67 MVA
Connection	YgD1
Primary winding resistance	0.003 pu
Primary winding inductance	0.06 pu
Secondary winding resistance	0.003 pu
Secondary winding inductance	0.06 pu
Iron-loss equivalent	500 pu
Magnetizing reactance	300 pu
Primary voltage level	69 kV
Secondary voltage level	22 kV

2) *Overhead line*: Table II shows the parameters of the block that simulates the zones; all such blocks share these parameters.

Table III shows the collected data for each section.

Finally, each zone is assigned a power factor of 0.86, which is used to determine its active and reactive power.

TABLE II
LINE PARAMETERS.

Parameter	Value
Positive-sequence resistance	13 mΩ/km
Zero-sequence resistance	390 mΩ/km
Positive-sequence inductance	0.94 mH/km
Zero-sequence inductance	4.13 mH/km
Positive-sequence capacitance	13 nF/km
Zero-sequence capacitance	8 nF/km

TABLE III
DATA FOR EACH ZONE.

Zone	Power	Length
Zone 1	2.5 MVA	14.2 km
Zone 2	3 MVA	10.2 km
Zone 3	6.1 MVA	20.7 km
Zone 4	1.5 MVA	49.1 km

IV. SIMULATION DEVELOPMENT AND RESULTS

After assembling the circuit of Fig. 1 in Simulink, the collected values are set through Matlab code to simplify the data handling. The Matlab code performs the necessary operations, mainly obtaining the load power values from the power factor. As a final step, voltage measurement points are placed at each element—the substation, the transformer primary and secondary, and each load. In the circuit, per-unit values are computed taking the base voltage upstream of the transformer as 69 kV and downstream as 22 kV.

To build the voltage profile, the last value of each measurement is taken, together with the distance at which it was obtained. The substation is taken to be at kilometer 0, the transformer primary at kilometer 1, and—to make the change visible—the transformer secondary at kilometer 3. For the zone distances, each zone length is added to the previous cumulative length. This gives Fig. 13, showing the voltage drop along the selected distance; the dashed lines indicate the kilometer at which each load is located.

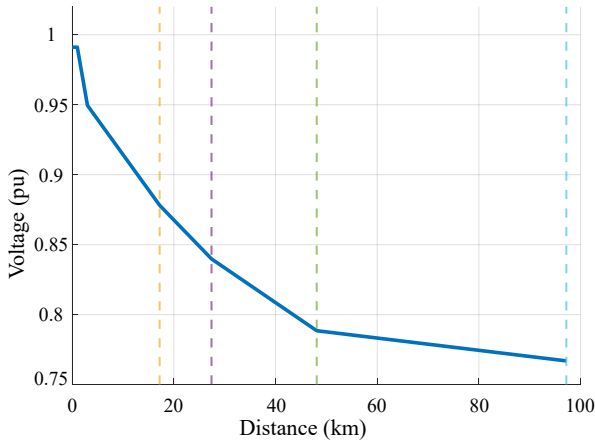


Fig. 13. Voltage drop along the selected overhead line - Case I

Table IV gives the specific data of the figure.

TABLE IV
SIMULATION RESULTS.

Element	Distance [km]	Voltage [pu]	Voltage drop [pu]
Substation	0	0.9911	—
Transformer primary	1	0.9911	0
Transformer secondary	3	0.9495	0.0416
Zone 1	17.2	0.8781	0.0714
Zone 2	27.4	0.8398	0.0383
Zone 3	48.1	0.7885	0.0513
Zone 4	97.2	0.7669	0.0216

The largest voltage drop from the transformer occurs in the transition from the secondary (ST) to zone 1 (Z1) and from zone 2 (Z2) to zone 3 (Z3), at 0.0714 and 0.0513 pu respectively. The secondary drop is not counted, since it is due to the transformer's own characteristic impedance and the values in Table I; ideally the per-unit voltage here should be 1.

The substation output and transformer primary are also not at 1 pu, which can be attributed to the generator impedance defined in the simulation ($X/R = 5$). Without this parameter the voltage would indeed be 1 pu, as shown in Fig. 14.

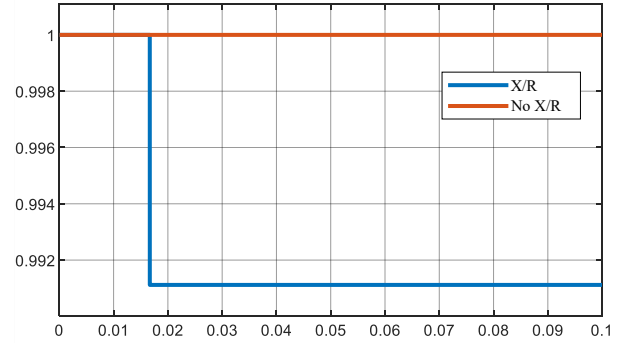


Fig. 14. Voltage difference with and without X/R .

Comparing Table IV with Table III, a considerable voltage drop accumulates by zone 4 due to the load and distance. Analyzing the zones in pairs, the largest drops occur between ST–Z1 and Z2–Z3, i.e., between zones that balance power and distance. The Z3–Z4 pair has the longest distance (49.1 km), but its load is only 1.5 MVA, giving a drop of 0.0216. The ST–Z1 span is 14.2 km with 2.5 MVA, while Z1–Z2 has a higher power (3 MVA) but a shorter distance (10.2 km). The Z2–Z3 drop is the most pronounced, since it combines the second-longest span (20.7 km) with the highest load (6.1 MVA).

One solution to this voltage drop is to install a voltage regulator in one of the zones. The problem statement requires simulating such a device if the selected line has one; the selected line does not, but to test its effectiveness, an imaginary regulator was placed in zone 4 with 80% of that zone's reactive load, simulated as a capacitor bank at kilometer 29.1 of zone 4. With these considerations, Fig. 15 shows a nearly stable voltage between kilometers 48.1 and 77.2 before it starts to drop again, already differing from Fig. 13.

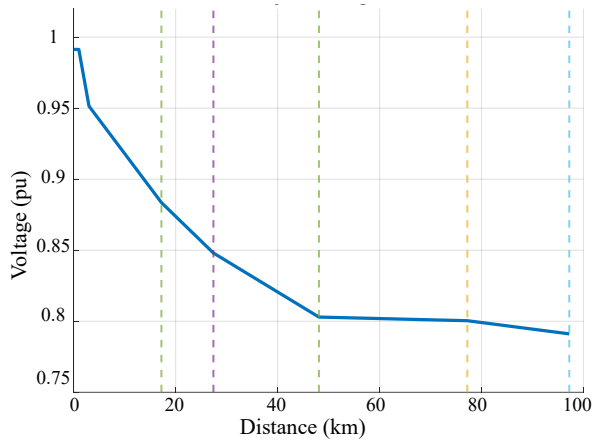


Fig. 15. Voltage drop along the line with a voltage regulator in zone 4 - Case I

Table V gives the corresponding data. The voltage drop is considerably reduced with respect to Table IV: the voltages are slightly higher, and at the end of zone 4 (nearly 97 km) there is a voltage rise of almost 0.03 pu compared with Table IV. A larger regulator placed closer to the first zones would yield an even higher voltage at the end of the section.

TABLE V
SIMULATION RESULTS WITH A VOLTAGE REGULATOR.

Element	Distance [km]	Voltage [pu]	Voltage drop [pu]
Substation	0	0.9914	—
Transformer primary	1	0.9914	0
Transformer secondary	3	0.9514	0.04
Zone 1	17.2	0.8836	0.0678
Zone 2	27.4	0.848	0.0356
Zone 3	48.1	0.8029	0.0451
Regulator zone	77.2	0.8004	0.0025
Zone 4	97.2	0.7911	0.0093

V. CONCLUSIONS

For this study, the Yanzatza substation and the Paquisha feeder—both in the eastern (Oriente) region of Ecuador—were selected, using the EERSSA technical geoportal and Google Maps for data collection. Matlab/Simulink proved to be an effective tool for analyzing and visualizing the voltage behavior along a distribution line and for identifying the voltage drop in each zone.

The analysis confirmed the influence of line distance and connected load on the voltage drop: from the transformer output at 0.9495 pu, the voltage falls to 0.7669 pu over the nearly 98 km of the selected line. The longest and most heavily loaded spans cause the most pronounced drops, as seen in the transitions from the transformer secondary to zone 1 and from zone 2 to zone 3.

The balance between a zone's distance and load is a critical factor reflected in the voltage profile. This is confirmed by the Z3–Z4 drop: this zone is the longest, but its low connected

load makes the drop relatively small, unlike zone 3, where distance and installed load are balanced and produce a larger drop.

The characteristic impedance of the substation (source) also matters: the internal source impedance, modeled through the X/R parameter, means the voltage between source and transformer is not exactly 1 pu. Since the transformer primary is not at 1 pu, the transformation to the secondary is not exactly 22 kV, which affects the voltages reaching the other zones—although this effect is minor, since the per-unit variation between substation and transformer is very small and some deviation from the ideal value is permitted in terms of service quality.

The results underscore the importance of considering both load distribution and zone distances in the design and planning of distribution systems to minimize voltage drops and improve service quality. In this case, zone 4 receives only 76.69% of the nominal voltage, which represents deficient service. Among the possible solutions, future work will simulate a transformer with taps, configured to deliver a higher voltage at the start of the line that falls to acceptable values along its length.

Another solution, simulated here, is a voltage regulator, whose role was to inject reactive power to raise the voltage at its location, delivering a nearly constant voltage over the rest of the section. Although the voltage drop is unavoidable, the regulator produced a per-unit voltage rise before the drop continued, an effective mitigation. It was placed in the longest zone, where the voltage reduction from the transformer output is most significant—which in practice would be the optimal location.

This work helped clarify how the voltage behaves from the substation to the various loads along the coverage area, confirming the theoretical knowledge from the course. Future work will evaluate a tap-changing transformer for the voltage profile and observe the system behavior when one of these loads is disconnected.

REFERENCES

- [1] Agencia de Regulación y Control de Energía y Recursos Naturales No Renovables, "Atlas del sector eléctrico ecuatoriano 2021," 2021.
- [2] Empresa Eléctrica Regional del Sur S.A. (EERSSA), "Plan de expansión del sistema eléctrico 2015–2025, Gerencia de Planificación," 2013.